

David Darling
4/23/26

HW #9

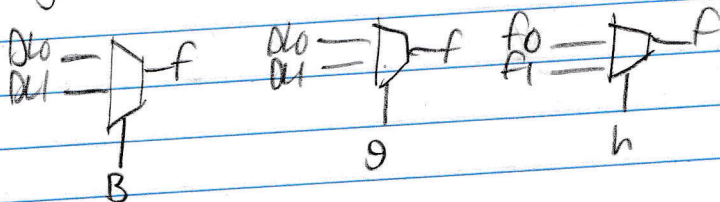
9.14 $A\bar{h}(a\bar{b} + bc) + h(e\bar{g} + fg)$

$f(w) = \bar{b}(a) + b(c)$

$f(v) = \bar{g}(e) + g(f)$

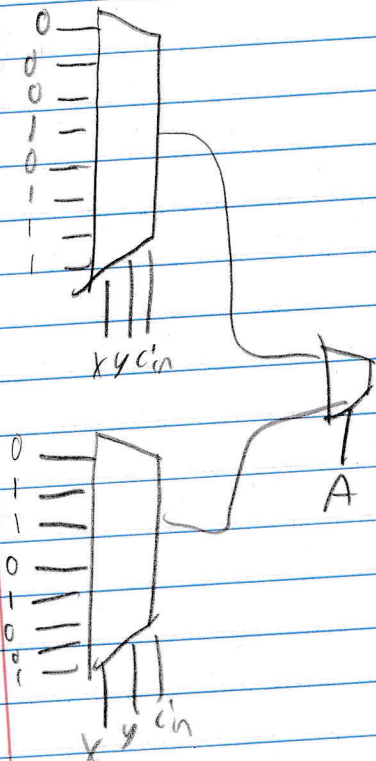
input $D_0 = a\bar{b} + bc$

input $D_1 = e\bar{g} + fg$



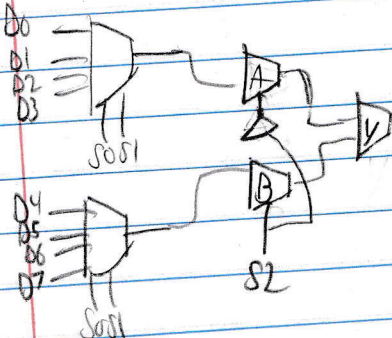
9.21

A)



A	B	C	Cin	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	0	1

9.25

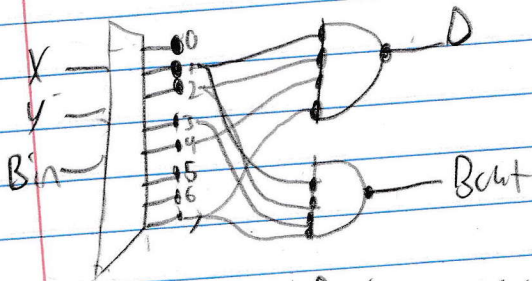


9.26 A)

A	B	B'	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$D = (\overline{m_1} \cdot \overline{m_2} \cdot \overline{m_4} \cdot \overline{m_7})' = m_1 + m_2 + m_4 + m_7$$

$$Bout = (\overline{m_1} \cdot \overline{m_2} \cdot \overline{m_3} \cdot \overline{m_7})' = m_1 + m_2 + m_3 + m_7$$



9.29 A)

RS74	Decimal	vw	yz
0000	0	00	00
0001	1	00	01
0010	2	00	10
0011	3	00	11
0100	4	01	00
0101	5	01	01
0110	6	01	10
0111	7	01	11
1000	8	10	00
1001	9	10	01
1010-1111	Invalid	xx	xx

